

TinyLogic UHS Triple Buffer

NC7NZ34

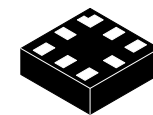
Description

The NC7NZ34 is a triple buffer from onsemi's Ultra High Speed Series of TinyLogic in the space saving US8 package. The device is fabricated with advanced CMOS technology to achieve ultra high speed with high output drive while maintaining low static power dissipation over a very broad V_{CC} operating range. The device is specified to operate over the 1.65 V to 5.5 V V_{CC} range. The inputs and outputs are high impedance when V_{CC} is 0 V. Inputs tolerate voltages up to 5.5 V independent of V_{CC} operating voltage.

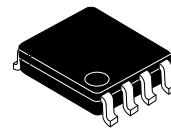
Features

- Space Saving US8 Surface Mount Package
- MicroPak™ Pb-Free Leadless Package
- Ultra High Speed: t_{PD} 2.4 ns Typ into 50 pF at 5 V V_{CC}
- High Output Drive: ± 24 mA at 3 V V_{CC}
- Broad V_{CC} Operating Range: 1.65 V to 5.5 V
- Power Down High Impedance Inputs / Outputs
- Overvoltage Tolerant Inputs Facilitate 5 V to 3 V Translation
- Proprietary Noise / EMI Reduction Circuitry Implemented
- These Devices are Pb-Free, Halogen Free/BFR Free and are RoHS Compliant

MARKING DIAGRAMS



UQFN8
1.6X1.6, 0.5P
CASE 523AY



US8
CASE 846AN



P9, NZ34 = Specific Device Code
KK, L = 2-Digit Lot Run Traceability Code
XY, YW = 2-Digit Date Code Format
Z, A = Assembly Plant Code

ORDERING INFORMATION

See detailed ordering, marking and shipping information in the package dimensions section on page 6 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 6.

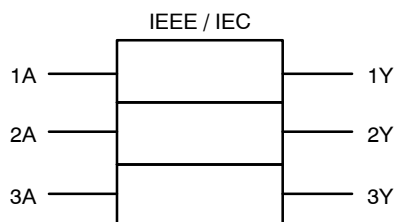


Figure 1. Logic Symbol

Connection Diagrams

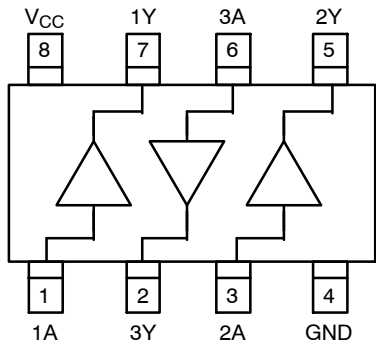


Figure 2. Connection Diagram (Top View)

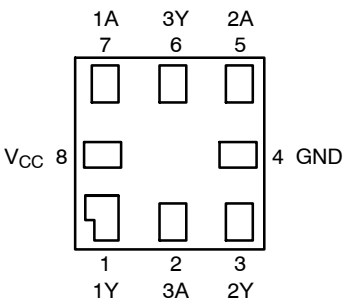
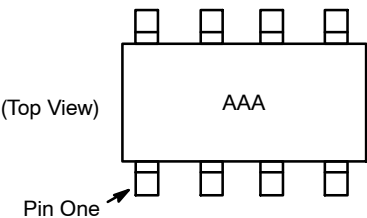


Figure 4. Pad Assignments for MicroPak (Top Thru View)



AAA represents Product Code Top Mark – see ordering code
NOTE: Orientation of Top Mark determines Pin One location.
Read the Top Product Code Mark left to right, Pin One is the lower left pin (see diagram).

Figure 3. Pin One Orientation Diagram

PIN DESCRIPTIONS

Name	Description
A ₁ , A ₂ , A ₃	Data Inputs
Y ₁ , Y ₂ , Y ₃	Output

FUNCTION TABLE (Y = A)

Input	Output
A	Y
L	L
H	H

H = HIGH Logic Level
L = LOW Logic Level

ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter		Min	Max	Unit
V _{CC}	Supply Voltage		-0.5	6.5	V
V _{IN}	DC Input Voltage		-0.5	6.5	V
V _{OUT}	DC Output Voltage		-0.5	6.5	V
I _{IK}	DC Input Diode Current	V _{IN} < 0 V	-	-50	mA
I _{OK}	DC Output Diode Current	V _{OUT} < 0 V	-	-50	mA
I _{OUT}	DC Output Source / Sink Current		-	±50	mA
I _{CC} / I _{GND}	DC V _{CC} / GND Current		-	±100	mA
T _{STG}	Storage Temperature		-65	+150	°C
T _J	Junction Temperature under Bias		-	+150	°C
T _L	Junction Lead Temperature (Soldering, 10 Seconds)		-	+260	°C
P _D	Power Dissipation in Still Air	US8	-	500	mW
		MicroPak-8	-	539	mW

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

RECOMMENDED OPERATING CONDITIONS

Symbol	Parameter		Min	Max	Unit
V _{CC}	Supply Voltage Operating		1.65	5.5	V
	Supply Voltage Data Retention		1.5	5.5	
V _{IN}	Input Voltage		0	5.5	V
V _{OUT}	Output Voltage		0	V _{CC}	V
t _r , t _f	Input Rise and Fall Time	V _{CC} = 1.8 V, 2.5 V ±0.2 V	0	20	ns/V
		V _{CC} = 3.3 V ±0.3 V	0	10	
		V _{CC} = 5.5 V ±0.5 V	0	5	
T _A	Operating Temperature		-40	+85	°C
θ _{JA}	Thermal Resistance	US8	-	250	°C/W
		MicroPak-8	-	232	°C/W

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

1. Unused inputs must be held HIGH or LOW. They may not float.

DC ELECTRICAL CHARACTERISTICS

Symbol	Parameter	V _{CC} (V)	Conditions	T _A = +25°C			T _A = -40 to +85°C		Unit
				Min	Typ	Max	Min	Max	
V _{IH}	HIGH Level Control Input Voltage	1.8 ±0.15		0.65 V _{CC}	–	–	0.65 V _{CC}	–	V
		2.3 to 5.5		0.7 V _{CC}	–	–	0.7 V _{CC}	–	
V _{IL}	LOW Level Control Input Voltage	1.8 ±0.15		–	–	0.35 V _{CC}	–	0.35 V _{CC}	V
		2.3 to 5.5		–	–	0.3 V _{CC}	–	0.3 V _{CC}	
V _{OH}	HIGH Level Control Output Voltage	1.65	V _{IN} = V _{IH}	I _{OH} = -100 µA	1.55	1.65	–	1.55	V
		2.3			2.2	2.3	–	2.2	
		3.0			2.9	3.0	–	2.9	
		4.5			4.4	4.5	–	4.4	
		1.65	V _{IN} = V _{IH}	I _{OH} = -4 mA	1.29	1.52	–	1.29	
		2.3		I _{OH} = -8 mA	1.9	2.14	–	1.9	
		3.0		I _{OH} = -16 mA	2.4	2.75	–	2.4	
		3.0		I _{OH} = -24 mA	2.3	2.62	–	2.3	
		4.5		I _{OH} = -32 mA	3.8	4.13	–	3.8	
V _{OL}	LOW Level Control Output Voltage	1.65	V _{IN} = V _{IL}	I _{OL} = 100 µA	–	0.0	0.1	–	V
		2.3			–	0.0	0.1	–	
		3.0			–	0.0	0.1	–	
		4.5			–	0.0	0.1	–	
		1.65	V _{IN} = V _{IL}	I _{OL} = 4 mA	–	0.08	0.24	–	
		2.3		I _{OL} = 8 mA	–	0.10	0.3	–	
		3.0		I _{OL} = 16 mA	–	0.16	0.4	–	
		3.0		I _{OL} = 24 mA	–	0.24	0.55	–	
		4.5		I _{OL} = 32 mA	–	0.25	0.55	–	
I _{IN}	Input Leakage Current	1.65 to 5.5	0 ≤ V _{IN} ≤ 5.5 V		–	–	±0.1	–	µA
I _{OFF}	Power Off Leakage Current	0.0	V _{IN} or V _{OUT} = 5.5 V		–	–	1.0	–	10 µA
I _{CC}	Quiescent Supply Current	1.65 to 5.5	V _{IN} = 5.5 V, GND		–	–	1.0	–	10 µA



AC ELECTRICAL CHARACTERISTICS

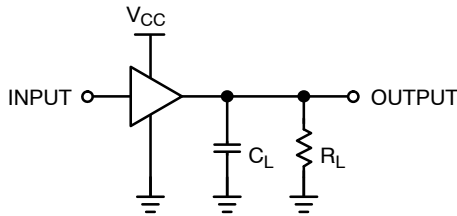
Symbol	Parameter	V _{CC} (V)	Conditions	T _A = +25°C			T _A = -40 to +85°C		Unit
				Min	Typ	Max	Min	Max	
t _{PLH} , t _{PHL}	Propagation Delay (Figure 5, 7)	1.8 ±0.15	C _L = 15 pF, R _L = 1 MΩ,	–	4.6	8.0	–	8.8	ns
		2.5 ±0.2		–	3.0	5.2	–	5.8	
		3.3 ±0.3		–	2.3	3.6	–	4.0	
		5.0 ±0.5		–	1.8	2.9	–	3.2	
		3.3 ±0.3	C _L = 50 pF, R _L = 500 Ω,	1.2	3.0	4.6	–	5.1	
		5.0 ±0.5		0.8	2.4	3.8	–	4.2	
C _{IN}	Input Capacitance	0		–	2.5	–	–	–	pF
C _{PD}	Power Dissipation Capacitance (Figure 6)	3.3	(Note 2)	–	9	–	–	–	pF
		5.0		–	11	–	–	–	

2. C_{PD} is defined as the value of the internal equivalent capacitance which is derived from dynamic operating current consumption (I_{CCD}) at no output loading and operating at 50% duty cycle. (See Figure 6). C_{PD} is related to I_{CCD} dynamic operating current by the expression: I_{CCD} = (C_{PD}) (V_{CC}) (f_{IN}) + (I_{CC}static).

AC ELECTRICAL CHARACTERISTICS

Symbol	Parameter	Conditions	V _{CC} (V)	T _A = +25°C	Unit
				Typical	
V _{OLP}	Quiet Output Dynamic Peak V _{OL}	C _L = 50 pF, V _{IH} = 5.0 V, V _{IL} = 0 V	5.0	0.8	V
V _{OLV}	Quiet Output Dynamic Valley V _{OL}	C _L = 50 pF, V _{IH} = 5.0 V, V _{IL} = 0 V	5.0	–0.8	V

AC Loading and Waveforms



C_L includes load and stray capacitance
Input PRR = 1.0 MHz, t_W = 500 ns.

Figure 5. AC Test Circuit

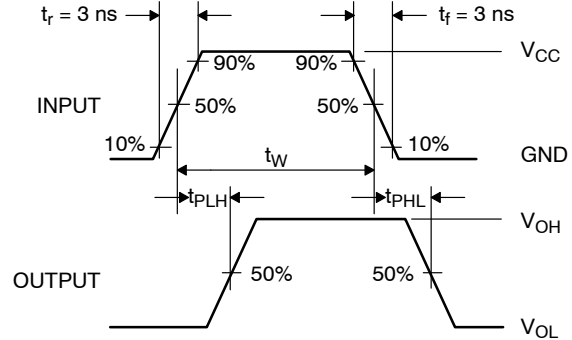
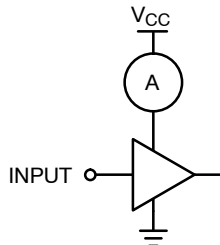


Figure 7. AC Waveforms



Input = AC Waveform; t_r = t_f = 1.8 ns;
PRR = 10 MHz; Duty Cycle = 50%.

Figure 6. I_{CCD} Test Circuit

NC7NZ34

ORDERING INFORMATION

Part Number	Top Mark	Package	Shipping [†]
NC7NZ34K8X	NZ34	8-Lead US8, JEDEC MO-187, Variation CA 3.1 mm Wide	3000 / Tape & Reel
NC7NZ34L8X	P9	8-Lead MicroPak, 1.6 mm Wide (Pb-Free)	5000 / Tape & Reel

DISCONTINUED (Note 4)

NC7NZ34K8X-L22236	NZ34	8-Lead US8, JEDEC MO-187, Variation CA 3.1 mm Wide	3000 / Tape & Reel
NC7NZ34L8X-L22185	P9	8-Lead MicroPak, 1.6 mm Wide (Pb-Free)	5000 / Tape & Reel

[†]For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

3. Pb-Free package per JEDEC J-STD-020B.

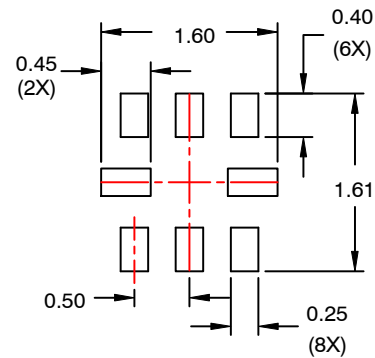
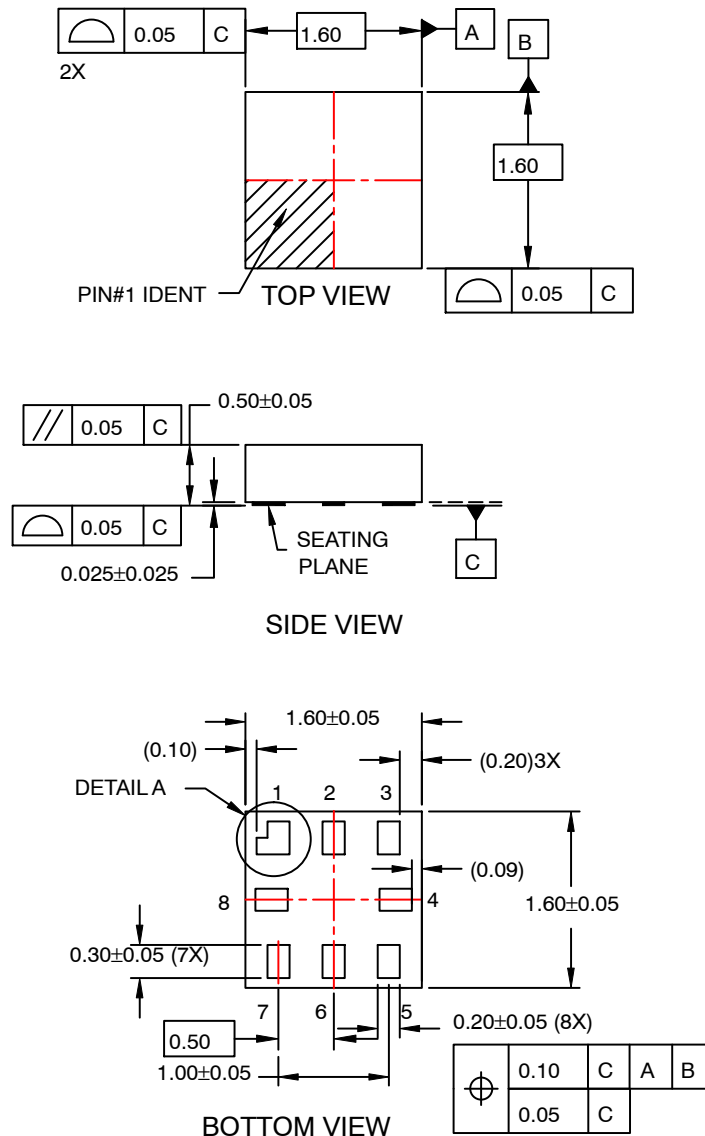
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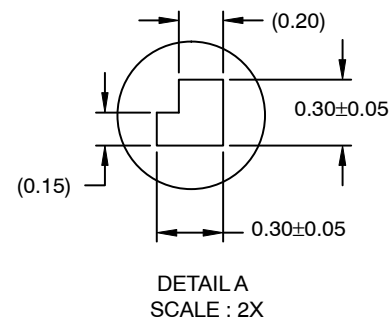
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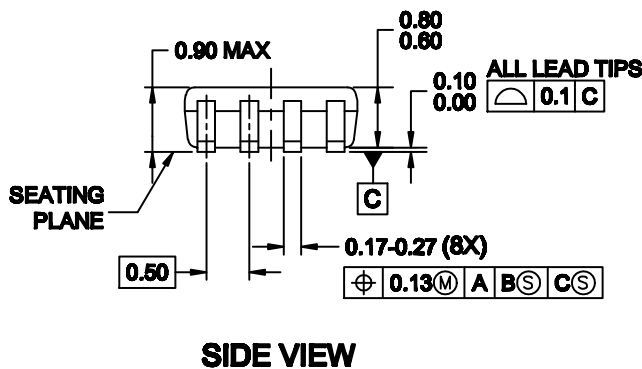
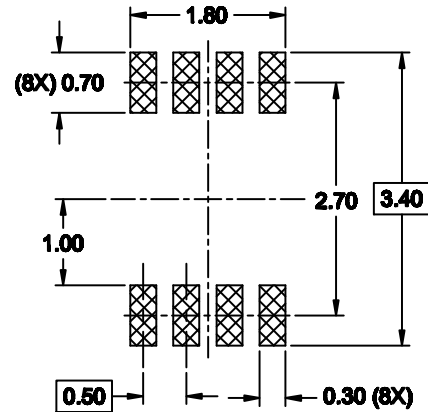
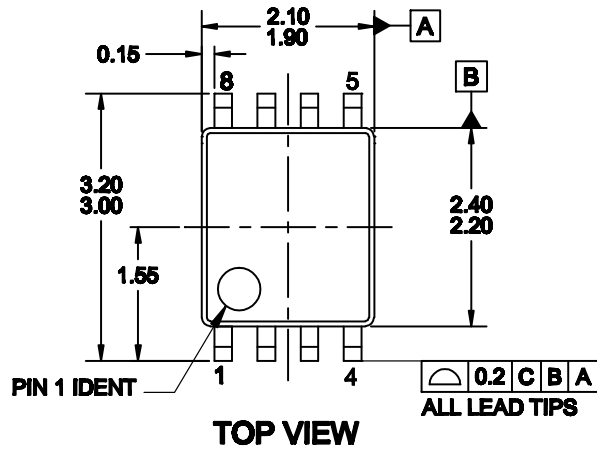


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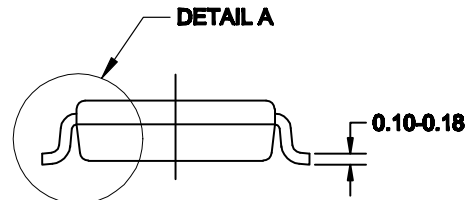
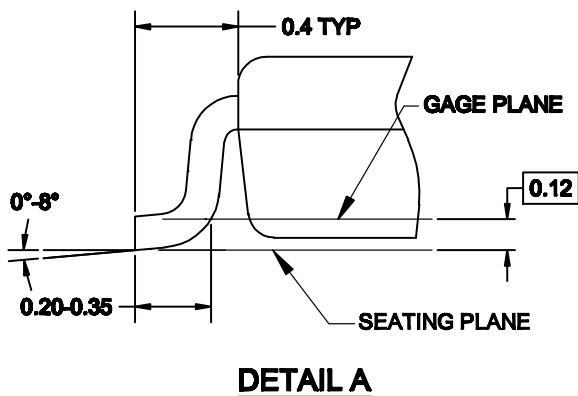
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